# Pipeline scalability with Airflow

In the realm of data processing and workflow management, mastering Apache Airflow is essential for designing and operating complex data workflows. In this lesson, we'll delve into various functionalities and configurations within Apache Airflow, from basic workflow creation to advanced integration and optimisation techniques allowing us to efficiently manage sophisticated data processing tasks.



### What You'll Learn:

**Identify and Utilise Airflow Operators**

> Identify and utilise various Airflow operators to design effective data workflows tailored to specific requirements.

**Implement Task Dependencies in Airflow**

> Understand and implement task dependencies within Airflow, employing strategies for linear, parallel, and branching task configurations.

**Configure and Manage Secure Connections**

Configure and manage secure connections in Airflow, establishing reliable links with external systems such as AWS and PostgreSQL databases.

**Use Hooks with the Python Operator** ✖

Learn to use hooks with the Python operator within Airflow to enhance functionality and tackle complex tasks that standard operators cannot handle.

**Analyse and Troubleshoot DAGs with Hooks and Operators** ✖

Analyse the setup and outcomes of DAGs that utilise hooks and operators, assessing their effectiveness and troubleshooting as needed.